

BAKANAS, Kazakhstan

WMO: 368210

Lat: 44.8117N

Lon: 76.2539E

Elev: 1299

StdP: 14.02

Time Zone: 6.00 (E06)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.6	-11.5	-22.5	1.7	-17.3	-16.6	2.3	-11.2	12.9	27.5	11.1	25.5	1.0	20	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	26.1	99.0	67.3	95.9	65.8	93.1	65.1	70.3	90.1	68.4	88.6	67.1	86.9	4.2	340

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	95.6	76.1	62.4	88.6	74.8	60.5	82.5	74.4	35.1	89.2	33.5	88.1	32.3	87.0	82.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.5	11.9	9.9	DB	-20.6	103.9	7.3	2.9	-25.8	105.9	-30.1	107.6	-34.2	109.2	-39.5	111.3
			WB	-20.9	72.6	7.2	3.4	-26.1	75.0	-30.3	77.0	-34.4	78.9	-39.6	81.4

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.4	14.6	20.9	40.3	54.4	64.7	74.9	77.7	74.5	63.6	50.1	35.6	19.8	(6)
(7)		DBStd	23.86	14.03	13.25	10.42	8.82	7.23	5.85	5.47	5.85	8.06	8.89	10.79	12.26	(7)
(8)		HDD50	3795	1099	815	329	56	1	0	0	4	108	445	938		(8)
(9)		HDD65	6870	1564	1235	767	326	96	5	1	3	120	466	884	1403	(9)
(10)		CDD50	3575	0	0	27	189	456	746	860	760	413	112	12	0	(10)
(11)		CDD65	1175	0	0	0	9	86	301	395	299	79	5	1	0	(11)
(12)		CDH74	15991	0	0	8	280	1352	3925	5219	3908	1193	103	3	0	(12)
(13)		CDH80	7866	0	0	0	73	503	1977	2824	1989	478	22	0	0	(13)
(14)	Wind	WSAvg	3.1	2.4	2.7	3.9	3.8	3.9	3.4	3.3	3.1	2.8	2.6	2.6		(14)
(15)	Precipitation	PrecAvg	7.8	0.6	0.6	0.6	0.6	0.9	0.7	0.8	0.4	0.3	0.8	0.9	0.6	(15)
(16)		PrecMax	12.4	1.9	1.5	1.7	1.7	2.0	2.3	1.8	1.8	1.9	2.1	2.5	1.2	(16)
(17)		PrecMin	4.1	0.1	0.1	0.2	0.0	0.1	0.0	0.1	0.0	0.0	0.0	0.1	0.2	(17)
(18)		PrecStd	2.2	0.4	0.4	0.4	0.4	0.4	0.6	0.5	0.4	0.4	0.6	0.6	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.2	50.8	74.3	87.1	94.5	100.1	104.0	101.7	95.1	83.4	68.9	48.1	(19)
MCWB			38.3	42.5	54.5	61.5	64.7	67.7	68.5	68.7	64.5	59.8	53.0	40.5	(20)	
2%		DB	38.9	44.4	66.8	81.1	89.5	96.5	99.9	97.7	90.2	76.4	60.9	41.4	(21)	
		MCWB	35.3	37.9	51.5	58.3	62.6	66.0	68.0	66.5	62.5	56.3	50.0	37.0	(22)	
5%		DB	35.4	40.9	62.1	76.7	85.6	93.6	96.3	94.2	85.7	70.9	54.9	38.0	(23)	
		MCWB	32.8	36.1	49.4	56.6	61.2	65.2	66.3	65.3	60.3	53.4	46.7	34.8	(24)	
10%		DB	32.5	37.2	57.1	71.6	81.9	90.6	93.1	90.5	81.2	66.1	49.6	35.0	(25)	
		MCWB	30.6	33.8	47.4	54.7	59.8	64.4	65.5	64.4	58.7	51.5	43.7	32.4	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	39.2	43.7	56.0	63.4	67.0	71.4	73.8	73.5	66.3	60.7	54.6	41.4	(27)
MCD B			42.6	48.8	69.9	79.8	86.9	90.7	94.4	94.0	91.4	79.9	65.7	46.1	(28)	
2%		WB	36.0	38.7	52.7	60.5	64.7	68.9	70.5	69.6	63.6	57.4	50.4	37.7	(29)	
		MCD B	38.5	43.4	64.8	76.6	83.8	88.5	90.3	90.2	85.7	72.1	58.7	41.0	(30)	
5%		WB	33.0	36.3	50.1	58.1	63.0	67.3	68.6	67.4	61.6	55.2	47.3	35.0	(31)	
		MCD B	35.1	40.2	60.4	72.6	80.5	86.8	88.9	87.9	82.3	68.1	54.1	37.8	(32)	
10%		WB	30.7	33.9	47.6	55.9	61.2	66.1	67.3	65.7	59.7	52.4	44.2	32.5	(33)	
		MCD B	32.5	37.1	56.8	69.1	77.8	85.2	87.0	86.1	78.8	64.2	49.2	34.9	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.9	17.9	20.4	25.0	27.4	26.5	26.1	28.1	28.1	24.0	17.4	15.4	(35)
MCD BR			17.6	18.9	28.6	33.9	32.7	30.9	31.5	33.0	34.2	31.6	25.0	17.4	(36)	
MCWBR			14.1	13.2	15.2	15.8	13.1	10.1	10.2	12.2	14.6	15.8	14.9	13.3	(37)	
5% WB		MCD BR	15.3	17.7	26.1	29.2	27.7	25.8	25.9	28.8	30.9	27.6	22.7	15.8	(38)	
		MCWBR	12.7	13.4	14.5	14.4	12.0	9.2	9.4	12.0	14.1	14.9	14.1	12.6	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.340	0.380	0.421	0.453	0.446	0.445	0.442	0.405	0.380	0.394	0.376	0.339	(40)	
(41)		taud	2.089	2.034	2.018	1.984	2.058	2.121	2.145	2.231	2.248	2.121	2.106	2.113	(41)	
(42)		Ebn at Noon	232	245	252	256	262	263	262	268	265	237	216	218	(42)	
(43)		Edh at Noon	32	41	48	54	51	49	47	41	37	37	31	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	532	847	1273	1753	2204	2318	2248	2068	1622	1020	593	406	(44)	
(45)		RadStd	61	72	86	116	97	99	130	87	64	79	44	40	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-2.12	-3.01	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	-1.39	-2.16	-365	-454	+250	+130	

Nomenclature: See separate page